

Fake News Detection using NER

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Abstract

Fake news often relies on name dropping public figures or institutions to boost credibility. This project explores whether named entity recognition (NER) can help automatically detect fake news by analyzing how names, places, and organizations are used in fake vs. real political statements. To do this, we use the LIAR2 dataset and two different NER models - RoBERTa and SpaCy. We extract entities from political statements and compare how they're used in false vs. true claims. The goal is to find out if certain types of entities, or the way they're mentioned, can be used as clues for automated fake news detection.

1 Introduction

Fake news is everywhere, especially in politics. It is used to influence opinions, stir up conflict, and boost agendas, often by sounding official or credible. One common strategy is name dropping public figures, locations, and institutions to make a statement seem more believable. This makes named entities an interesting topic to explore when trying to detect misinformation automatically.

This project looks at how NER can help distinguish between real and fake political statements. If certain types of entities, like nationalities, organizations, or numbers appear in one class more than in the other, or if they're used differently depending on the truthfulness of the statement, that could be a useful indicator. We focus on analyzing entity patterns and what they reveal.

We use the LIAR2 dataset (Table 3), which contains short political claims labeled with truth values, and apply two different NER models to it - RoBERTa and SpaCy. These were chosen because they represent two different approaches to NER. RoBERTa is a transformer model that uses deep contextual understanding but has a relatively limited label set. On the other hand, SpaCy combines statistical and rule based components and provides more varied entity types, like `NORP` or `PERCENT`. Comparing them lets us see how model architecture and label design affect the kinds of patterns we can pick up.

The main goal is to find out whether the presence, frequency, or type of named entities can be linked to fake vs. real statements. Along the way, we also look at how wrapping the entities in their labels influences the outcome. This could help future work in automated fake news detection by showing whether entities are a meaningful feature to focus on, and if so, what kind of NER system works best.

1.1 Related Work

Fake news detection has been widely studied using a variety of features and modeling techniques. (Wang, 2017) introduced the LIAR dataset and applied machine learning models such as logistic regression, SVMs, CNNs and Bi-LSTMs to classify political claims. Their approach relied on linguistic features and speaker metadata, using a hybrid CNN architecture that combined textual content with contextual features like party affiliation to improve performance. This work highlighted the importance of metadata in detecting misinformation.

More recent work has explored the use of NER as a part of explainable fake news de-

tection systems. (De Magistris et al., 2022) proposed a system that uses NER to extract relevant sentences and retrieve evidence from trusted sources. Rather than predicting truthfulness directly, their system uses a stance classifier to evaluate how retrieved documents relate to the input claim, aiming for greater transparency and user interpretability.

Our work differs from both of these approaches. We take a simpler, more focused route. We use NER not to retrieve documents or model stance, but as a source of direct input features. Specifically, we extract the types and counts of named entities for each claim and use them in machine learning models to predict whether the statement is true or false. We also explore whether explicitly marking entities in the text using XML tags (`<PERSON>`, `<ORG>`) improves model performance. Unlike (Wang, 2017), we do not use speaker or contextual metadata, and unlike (De Magistris et al., 2022), we do not rely on external document retrieval. Our focus is on how entities function within the statements themselves.

1.2 Research Question

To explore this topic, we divided our research into two main research questions:

- Can classification models predict the truthfulness of a statement using only the counts of named entity types, as extracted by different NER systems?
- Does adding XML tags around named entities in the input text help improve fake news classification performance across different NER systems?

These questions test different ways of using NER for fake news detection. The first one focuses on what entities are mentioned and how often. They're encoded as features and fed into different machine learning models to see how well they perform. The second goes further and checks if explicitly marking entities in the text helps models understand the content better. Comparing both tells us if the extra complexity actually improves results or not.

2 Data

We used the LIAR2 dataset 3, which contains political statements accompanied by a "truthfulness" score ranging from 0 to 5. Scores of 0, 1, and 2 indicate varying degrees of falseness, while scores of 3, 4, and 5 represent varying degrees of truthfulness. The dataset also includes additional background information about the speaker, such as the number of previous true or false statements and the context in which the statement was made - however, this information is not required for our analysis.

For preprocessing, we extracted only the relevant columns: the statement and its corresponding truthfulness label. We then converted the original multiclass label into a binary format, assigning 0 to false statements and 1 to true statements to simplify the task.

3 Methods

3.1 RoBERTa

We chose RoBERTa over other NLP models like BERT, GPT because it performs exceptionally well (Wang, 2017) on token classification tasks such as named entity recognition (NER). Compared to BERT, RoBERTa is trained on a larger dataset and with improved training techniques, which helps it better capture context in short, dense texts. For our use case—analyzing political statements in the LIAR dataset—this ability to model subtle language patterns makes RoBERTa a strong choice. Models like GPT, while powerful for generation, are less suited for structured tasks like NER.

When fine-tuned for named entity recognition, RoBERTa labels each token in the context of the entire sentence. This allows it to resolve ambiguities more effectively than models that rely only on local context. However, a limitation of RoBERTa in this task is its relatively small and coarse-grained label set, typically including only categories like PER, ORG, LOC, and MISC.

For example, in the sentence: *"90 percent of Americans support universal background checks for gun purchases."* RoBERTa produces the following output:

"Americans" → MISC

While the model correctly identifies "Americans" as an entity, the assigned label is broad due to the limited tag set. Despite this, we chose RoBERTa to investigate whether strong contextual representations can still provide useful signals for fake news detection, even when entity categories are not very specific.

3.2 SpaCy

SpaCy is a popular NLP library with a well-defined entity label set, including specific types like percentages and nationalities. It processes text through a rule-based pipeline combined with statistical models.

On the same sentence, SpaCy outputs:

→ "90 percent" → PERCENT

→ "Americans" → NORP

Unlike RoBERTa, SpaCy distinguishes both the quantity ("90 percent") and the nationality ("Americans") with precise labels. SpaCy uses a larger set of entity labels, allowing it to distinguish specific types like nationalities (NORP) or percentages (PERCENT), whereas RoBERTa often uses broader categories such as MISC or LOC. SpaCy's richer tag set allows for more nuanced interpretation of entities, which can be useful for certain tasks.

We included SpaCy to see whether a lighter and faster model trained on significantly less data could perform competitively. It serves as a useful baseline for comparing performance-efficiency trade-offs. SpaCy's specificity in labeling and computational efficiency make it a strong candidate for contrast with transformer-based models like RoBERTa.

4 Experiments

4.1 Part 1: Entity Feature Extraction and Classification

In this part, we built simple prediction models for fake news detection using pretrained NER models. We started by using the LIAR2 dataset, which contains labeled political statements, and performed some basic preprocessing, such as combining the labels into a binary label and dropping unnecessary features.

First, we extracted entity types from each statement using a pretrained NER model

(roBERTa, SpaCy) and stored them as raw features. Then, we processed these entities by listing their types (like PERSON, DATE, PERCENT, etc.) and created a hot encoded dataset, where each statement was represented by a vector indicating which entity types appeared, and how many times they appeared. After that, we added the binary labels back into the dataset to prepare it for machine learning. Using these features, we trained several ML models (Logistic Regression, SVM, KNN, Decision Tree) to predict whether a statement was true or false. Finally, we compared the model performances against each other and against a random baseline to evaluate how much using entity features helped.

4.2 Part 2: Evaluating the Impact of XML-Tagged Entities on Text Classification

In this part, we investigated whether injecting XML tags into the text to highlight entity labels could improve the performance of our fake news classification models. The goal was to make named entities more salient to the model by wrapping them in tags such as `<person>Obama</person>`, potentially helping the model learn more entity-aware representations.

We conducted this experiment using both BERT and SpaCy classifiers, in two settings:

- Without XML: trained on raw statements only
- With XML: trained on statements where named entities were wrapped with entity-specific XML tags

BERT with XML tags We used a fine-tuned `bert-base-uncased` model for binary classification. Before training, we inserted XML tags into the text using pre-extracted entities from RoBERTa, merged adjacent entities, and tokenized the modified text. We also extended the tokenizer with additional special tokens for the XML tags (e.g., `<per>`, `</per>`, etc.), so that the model could treat them as distinct input tokens. We performed grid search on learning rate, number of epochs

and batch size using training data and evaluating on validation data. Parameters returning best F1 score were used in final BERT training, where the model was trained on both training and validation data and evaluated on the test dataset.

SpaCy with XML tags We also trained SpaCy’s `TextCatBOW.v3` model using the XML-tagged input. `TextCatBOW.v3` is SpaCy’s classifier based on a Bag-Of-Words approach (hence BOW), where each input is represented as a vector that counts the occurrences of individual tokens, and is then passed through a neural network to predict class labels. Each entity annotated sentence was paired with its binary label using SpaCy’s expected format.

We trained for 10 epochs with a batch size of 8, applying random shuffling each epoch. Training examples were wrapped in `Example` objects to align the text and annotations. As with BERT, we tracked accuracy, F1, precision, and recall on both validation and test sets.

5 Results

5.1 Part 1: Entity Feature Extraction and Classification

We evaluated the performance of four machine learning models trained on entity type count features extracted using two different NER systems: RoBERTa and SpaCy. Each input was represented as a vector indicating the count of named entity types in the statement. We report accuracy, F1-score, precision, and recall for each combination of NER model and ML classifier. We also compare against a random baseline.

Model (NER source)	Accuracy	F1-score	Precision	Recall
Random Baseline	53.31%	44.89%	44.71%	44.80%
LogReg (SpaCy)	61.76%	61.20%	61.76%	58.41%
SVM (SpaCy)	61.98%	61.22%	61.98%	59.57%
KNN (SpaCy)	59.15%	58.02%	59.15%	57.76%
Decision Tree (SpaCy)	60.89%	59.87%	60.89%	58.64%
LogReg (RoBERTa)	57.62%	33.20%	57.62%	42.13%
SVM (RoBERTa)	56.88%	55.45%	56.88%	55.29%
KNN (RoBERTa)	55.62%	52.90%	55.62%	51.80%
Decision Tree (RoBERTa)	56.62%	56.15%	56.62%	56.31%

Table 1: Performance of ML classifiers trained on entity count features extracted with SpaCy and RoBERTa

From the Table 1 and Figures 1 and 2, we will be able to assess whether entity types extracted using SpaCy or RoBERTa offer more discriminative signal, and whether simple models like Logistic Regression or Decision Trees are sufficient for this task.

5.2 Part 2: Evaluating the Impact of XML-Tagged Entities on Text Classification

In this section, we present the performance of our fake news classifiers trained with and without XML-tagged entity annotations. We compare models across four key metrics: Accuracy, F1-score, Precision, and Recall. Our goal is to determine whether adding explicit NER-based structure improves classification.

Model	Accuracy	F1-score	Precision	Recall
Random Baseline	50.00%	49.80%	50.10%	49.50%
SpaCy (no XML)	67.03%	60.68%	61.34%	60.02%
SpaCy (with XML)	65.90%	59.32%	59.98%	58.68%
BERT (no XML)	69.68%	64.98%	63.64%	66.39%
BERT (with XML)	68.55%	63.79%	62.29%	65.36%

Table 2: Performance comparison of models on the test set with and without XML-tagged entities

As shown in Table 2, both SpaCy and BERT classifiers achieve a substantial performance gain over the random baseline, highlighting the effectiveness of using pretrained language models for fake news classification. Among all models, **BERT without XML tagging** achieves the highest F1-score (64.98%) and overall accuracy (69.68%), indicating its strong ability to capture contextual signals in the text.

Interestingly, adding XML-tagged entities leads to a slight decline in performance for both SpaCy and BERT models. For instance, the F1-score of BERT drops from 64.98% to 63.79% when XML annotations are included. This suggests that, while entity information may provide additional structure, it could also introduce noise or disrupt the model’s internal representation learned during pretraining.

In conclusion, while the use of XML-tagged entity annotations is conceptually appealing, our results suggest that current implementations do not consistently enhance performance.

6 Analyses

6.1 Part 1: Entity Feature Extraction and Classification

To understand which named entity types contribute most to fake news detection, we conducted an ablation study using Logistic Regression trained on different combinations of NER tag count features (PER, ORG, LOC, MISC)(Table 4). The NER tags were extracted using RoBERTa. Logistic Regression is a simple but effective linear model that allows us to assess how much each feature contributes to the prediction without added complexity.

We observe that using all four entity types provides the most balanced performance overall. Interestingly, combinations like PER, LOC and even single features like ORG result in high recall, suggesting some entity types may carry strong signals for falsehoods. However, their lower precision shows that these signals are not always reliable on their own.

6.2 Part 2: Evaluating the Impact of XML-Tagged Entities on BERT Text Classification

For the purpose of understanding the behavior of BERT model with and without XML-tags, we tried to compare the errors (Figure 3).

In cases where XML-tagged BERT model underperformed, tags could have negatively impacted the prediction. The model was likely overweighting entities PER, ORG and LOC, which includes names of organizations, countries and famous people. This led to failing at grasping the whole context and therefore making incorrect predictions.

XML-tagged model prevailed often when the statements had structured numerical data. The input was possibly assembled better and pinpointing the impactful information was easier.

Barplot shows how frequent each entity type was in misclassification. Entity type PER was the most dominant tag for both models, suggesting that highlighting well-known individuals may have led the models to incorrectly classify certain claims. However, since PER is also the most frequent tag overall, its dominance in errors may simply reflect its overall frequency.

7 Discussion

Our project showed that NER can help with fake news detection, but its impact depends on the method. In Part 1, SpaCy slightly outperformed RoBERTa, likely due to its more detailed entity labels. SVM and Logistic Regression reached around 62% accuracy, beating the random baseline, showing that even simple models can use entity-based features effectively. However, the overall performance was modest, suggesting entity counts alone aren't enough.

In Part 2, wrapping entities in XML tags slightly worsened results. Models trained on raw text performed better, likely because the tags added noise. Tags may have caused models to overemphasize entities like PER, leading to misclassification. This disproves our hypothesis that tagging would improve accuracy.

8 Conclusion

Our project explored two NER-based approaches for fake news detection: using entity counts and highlighting entities with XML tags. We found that detailed tagging (like SpaCy's) helped simple models like SVM and Logistic Regression outperform a random baseline, but the gains were modest, and entity counts alone weren't highly predictive.

Tagging entities with XML in the input turned out less effective. Instead of helping, the tags often added noise and led models to overfocus on certain names or terms. Though logical in theory, this method didn't improve performance and may hurt results if not applied carefully.

9 Appendix

9.1 Group Contributions

- Writing - Lucie Navratilova, Matus Stanko, David Poda
- Data Exploration - Lucie Navratilova
- Everything RoBERTa/BERT (part 1 & part 2, code) - Matus Stanko
- Using HPC - Matus Stanko

- Everything SpaCy (part 1 & part 2, code) - Lucie Navratilova
- Error analysis - David Poda

9.2 ChatBot Usage

We used ChatGPT and DeepL to refine our writing, mainly for grammar, rephrasing, and synonym suggestions. Our process started with freely writing down ideas, then editing for structure and clarity, with AI tools used for final polishing. We also used them to find related literature and explore our hypothesis.

For coding, ChatGPT helped us fix bugs and understand unfamiliar libraries like RoBERTa and spaCy. We wrote the rest independently when working with tools we already knew.

9.3 Graphs, tables and figures

Label	Statement
1	90 percent of Americans “support universal background checks” for gun purchases.
0	Last year was one of the deadliest years ever for law enforcement officers.
0	Bernie Sanders’s plan is “to raise your taxes to 90 percent.”
1	Voter ID is supported by an overwhelming majority of NY-ers, from all across the state, walks of life, & political parties.
0	Says Barack Obama “robbed Medicare (of) \$716 billion to pay for ... Obamacare.”

Table 3: Sample statements with labels (Xu and Kechadi, 2024)

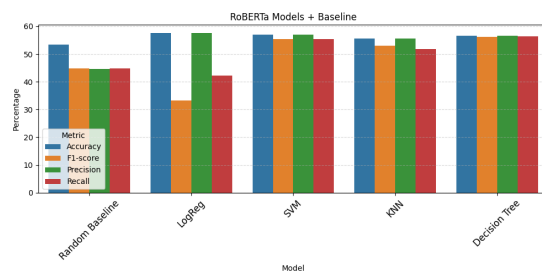


Figure 1: Performance of ML classifiers trained on RoBERTa tagged data

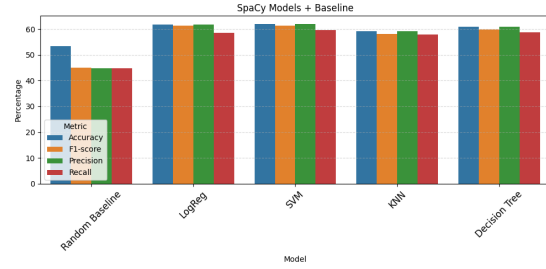


Figure 2: Performance of ML classifiers trained on SpaCy tagged data

Model	Accuracy	F1-score	Precision	Recall
PER, ORG, LOC, MISC	54.75%	54.17%	47.45%	63.10%
PER, LOC	53.01%	55.94%	46.41%	70.40%
PER only	52.87%	56.02%	46.33%	70.81%
ORG only	46.43%	56.44%	43.06%	81.91%
LOC only	52.44%	39.47%	42.84%	36.59%
MISC only	45.64%	52.40%	41.66%	70.61%

Table 4: Logistic Regression performance using individual or combined NER tag count features (RoBERTa-based)

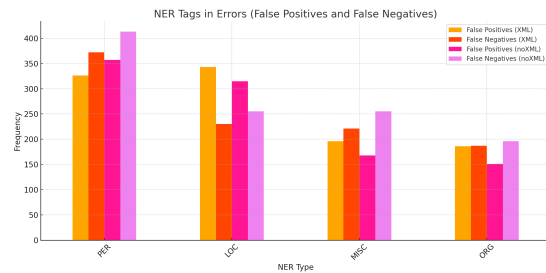


Figure 3: Frequency of NER tags in misclassified cases

References

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